Name: Will Hutter

P set #: 3

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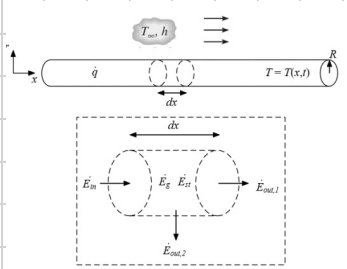
Problem 1

Known: R, $\dot{q}$, T_∞, h

Find:
a) Energy balance as a function of T
b) Governing equation to describe T profile
c) boundary conditions

Schematic:



Assumptions: neglect radiation, $R \ll L$, Radially uniform T
k, ρ, c_p constant

Analysis:

a)

$$\dot{E}_{in} + \dot{E}_g - \dot{E}_{out_1} - \dot{E}_{out_2} = \dot{E}_{st}$$

$$\dot{E}_{in} = -kA\frac{dT}{dx} \qquad \dot{E}_{out_1} = -kA\left(\frac{dT}{dx} + \frac{d^2T}{dx^2}\right) \qquad \dot{E}_{st} = mc_p\frac{dT}{dx} = \rho c_p(A\,dx)\frac{dT}{dx}$$

$$\dot{E}_g = \dot{q}A\,dx \qquad \dot{E}_{out_2} = hA_s(T - T_\infty)$$

$$\cancel{-kA\frac{dT}{dx}} + \cancel{kA\frac{dT}{dx}} + kA\frac{d^2T}{dx^2} - hA_s\cancel{dx}(T - T_\infty) = \rho c_p(A\cancel{dx})\frac{dT}{dx}$$

$$kA\frac{d^2T}{dx^2} - h\,2\pi R(T - T_\infty) + \dot{q}A = \rho c_p A\frac{dT}{dx}$$

b)

$$k\pi R^2\frac{d^2T}{dx^2} - h\,2\pi R(T - T_\infty) + \dot{q}\pi R^2 = \rho c_p \pi R^2\frac{dT}{dx}$$

$$k\frac{d^2T}{dx^2} - \frac{2h}{R}(T - T_\infty) + \dot{q} = \rho c_p\frac{dT}{dx}$$

c) $x = 0 \rightarrow T(0,t) = T_1 \quad (t > 0)$

$x = L \rightarrow \quad -k \frac{dT}{dx}\Big|_{x=L} = h\,(T(L,t) - T_\infty) \quad (t > 0)$

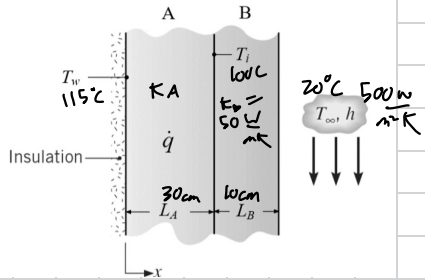
$t = 0 \rightarrow \quad T(x,0) = T_i \quad (0 \leq x \leq L)$

problem 2

Known: $L_A = 30\,cm$, $\dot{q}$, k_A, $L_B = 10\,cm$, $k_b = 30\,W/mK$
$T_\infty = 20°C$ $h = 500\,W/m^2K$ $T_w = 115°C$ $T_i = 100°C$

Find:
a) T_s on Right of B
b) $\dot{q}$ at steady state
c) k_A
d) plot temp profile across wall A + b

Schematic:



Assumptions: left A insulated, perfect contact, no heat gen in B
Uniform properties

Analysis:

a) $R_B = \frac{L_B}{k_B A_B} = \frac{T_i - T_s}{q}$ $q = hA(T_s - T_\infty)$

$\frac{L_B}{k_B A_B} = \frac{T_i - T_s}{hA(T_s - T_\infty)}$ $L_B hA(T_s - T_\infty) = (T_i - T_s) k_B A_B$

$T_s = (T_i - T_s)\frac{k_B A_B}{L_B hA} + T_\infty$

$T_s = \frac{k_B T_i + L_B h T_\infty}{k_B + L_B h}$

$= \frac{30(100) + 0.1 \cdot 500 \cdot 20}{30 \cdot 0.1 \cdot 500} = 50°C$

b) $$q''_A = \frac{T_s - T_i}{L_B} k_B = \frac{50-20}{0.1} \cdot 50 = 15000\ W/m^2$$

$$\dot{q} = \frac{15000\ W/m^2}{0.3\ m} = 50{,}000\ W/m^3$$

c) $$k_A \frac{d^2T}{dx^2} + \dot{q} = 0$$

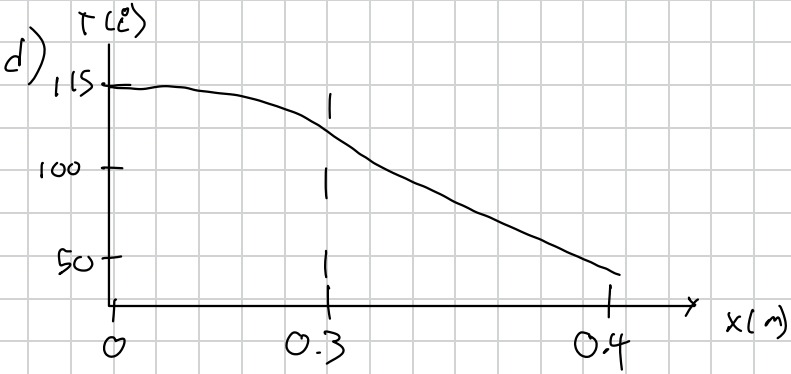
$$\frac{dT}{dx} = \frac{-\dot{q}}{k_A} x + \cancel{C} \qquad \left.\frac{dT}{dx}\right|_{x=0} = 0$$

$$T = \frac{-\dot{q}}{2k_A} x^2 + D \qquad T(0) = T_w$$

$$T = \frac{-\dot{q}}{2k_A} x^2 + T_w \qquad T(L_A) = T_i = -\frac{\dot{q}}{2k_A}(L_A)^2 + T_w$$

$$k_A = \frac{\dot{q} L_A^2}{2(T_w - T_i)} = 150 \frac{W}{m K}$$

d)



problem 3

Known: $k_b = 25\ W/mK$ $\quad k_{TBC} = 1\ W/mK$ $\quad R'' = 10^{-4}\ m^2 \cdot K/W$

$T_{\infty,i} = 400°K$ $\quad T_{\infty,o} = 1616\ K$ $\quad h_o = 1000\ W/m^2K$ hot

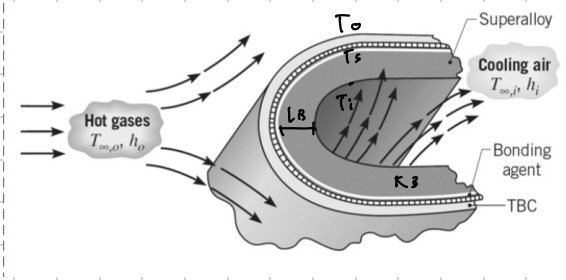
$h_i = 500\ W/m^2K$ cool

$L_b = 5\ mm$ $\quad T_{max} = 1200\ K$

Find: a) thermal res network and T_s

b) same w/ TBC applied

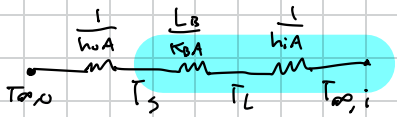
schematic:



Assumptions: const material properties, effectively flat

Analysis:

a)



$$R_{eq} = \frac{L_B}{k_BA} + \frac{1}{h_iA} \qquad \Delta T = q\,R_{eq}$$

$$q = (T_{\infty,o} - T_s)\,h_oA = \frac{T_s - T_{\infty,i}}{\frac{L_B}{k_BA} + \frac{1}{h_iA}}$$

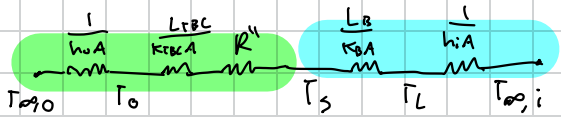
$$T_{\infty,o}(\sim) - T_s(\sim) = T_s - T_{\infty,i}$$

$$\sim = \frac{h_oA}{\frac{L_B}{k_BA} + \frac{1}{h_iA}} = 6.875 \cdot 10^{-4} \qquad T_s = \frac{T_{\infty,o}(\sim) + T_{\infty,i}}{1 + (\sim)}$$

$$= 1236\ K$$

Overheats

b)



$$AR_{eq,1} = \frac{L_{TBC}}{k_{TBC}\cancel{A}} + \frac{1}{h_0 A} + \frac{R''}{\cancel{A}} \qquad AR_{eq,2} = \frac{L_B}{k_B \cancel{A}} + \frac{1}{h_i \cancel{A}}$$

$$\Delta T = q\, R_{eq} \qquad q = \frac{\Delta T}{R_{eq}}$$

$$AR_{eq,1} = 0.0016 \qquad R_{eq,2} = 0.0022$$

$$\frac{\cancel{A}\, T_{\infty,0} - T_s}{A\, R_{eq,1}} = \frac{\cancel{A}\, T_s - T_{\infty,i}}{A\, R_{eq,2}}$$

$$(T_{\infty,0} - T_s) R_{eq,2} = R_{eq,1}(T_s - T_{\infty,i})$$

$$\frac{T_{\infty,0} R_{eq,2} + T_{\infty,i} R_{eq,1}}{R_{eq,1} + R_{eq,2}} = T_s = 1104°K$$

$< T_{max}$, cool enough